

Advaith Moholkar

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in [Advaith-Moholkar](#)

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Education

Indian Institute of Technology (IIT), Jodhpur

B.Tech. Engineering Science (Majors in CSE)

Aug. 2023 – May 2027

CGPA: 8.16/10

Jayshree Periwal High School, Jaipur

Senior Secondary

Apr. 2021 – May 2023

Percentage: 95.6/100

Technical Skills

- **Programming:** Python, C/C++, SQL (PostgreSQL, MySQL), JavaScript, HTML/CSS
- **ML/DL:** PyTorch, TensorFlow, scikit-learn, OpenCV, FinBERT
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Excel

Projects

LaneNet: Enhanced Curved Lane Detection

(Mentor - Dr. Himanshu Kumar)

[GitHub](#) 

Jan. 2025 - May 2025

- Developed **LaneNet**, a deep neural network for **learned label assignment**, integrated into **CLRNet** to create **CLRLaneNet**, achieving a **+2.8% F1 gain** on the **CULane dataset (100K+ images, 9 categories)**.
- Replaced static cost-based matching with a **fully connected encoder-decoder module**, enabling better generalization for curved lanes, shadows, and low-light scenarios.
- Trained on **8,677 curved-lane samples** using transfer learning and fine-tuned CLRNet weights, improving precision.
- **Tools & Technologies:** Python, PyTorch, CLRNet, ResNet, OpenCV, Albumentations, W&B.

Smart Product Pricing

(Amazon ML Challenge '25)

[GitHub](#) 

Oct. 2025

- Developed a **multimodal ML system** to predict e-commerce product prices using **textual and visual data**, achieving a **final SMAPE of 48.94**.
- Engineered a dual-CNN visual pipeline (**EfficientNet-B2, ConvNeXt-Tiny**) and a hybrid textual pipeline (**TF-IDF, SentenceTransformers, Regex-based features**).
- Implemented a **LightGBM ensemble** optimized via **weighted blending** and **5-fold cross-validation**.
- **Tools & Technologies:** Python, PyTorch, LightGBM, Hugging Face Transformers, NumPy, Pandas.

QuantSim: Quantitative Factor Investing Simulator

(Self Project)

[GitHub](#) 

May 2025 - Jun. 2025

- Developed a **multi-factor investment simulator** using **Random Forest models** and **FinBERT sentiment analysis**, achieving investor net **12.2% return (Sharpe 0.68)** on a 500-stock dataset (2010–2016).
- Implemented **risk-adjusted allocation** (inverse volatility) and an **API-ready NLP pipeline** for real-time sentiment scoring.
- Built a **Streamlit dashboard** for factor visualization, allocation insights, and portfolio performance monitoring.
- **Tools & Technologies:** Python, Pandas, NumPy, scikit-learn, FinBERT, Matplotlib, PostgreSQL, Streamlit.

Financial Risk Assessment

(Mentor - Dr. Avinash Sharma)

[GitHub](#) 

Oct. 2024 - Nov. 2024

- Built a **credit risk scoring system** using **Random Forest, XGBoost, SVM**, achieving **98.8% accuracy** with hyperparameter tuning and deployed a **Streamlit-based interface**.
- **Tools & Technologies:** Python, scikit-learn, Flask, Streamlit, Pandas, NumPy, Matplotlib.

Relevant Coursework

- Probability, Statistics and Stochastic Processes(A) ; Machine Learning(A) ; Linear Algebra and Calculus(A-) ; Design and Analysis of Experiments(A-) ; Data Structures and Algorithms(A-)

Achievements

- Ranked in the **Top 300** out of 82K+ participants in the **Amazon ML Challenge '25**.

Certifications

- Amazon ML Summer School '25
- Machine Learning Specialization - DeepLearning.AI

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